Sahaj Bhadja

(213) 284-4651 | bhadja@usc.edu | LinkedIn | GitHub | Portfolio

Summary

Full-stack software engineer with 2+ years building production **React**, **TypeScript**, and **Python** systems. Experienced in real-time services, concurrent processing, simulation workflows, distributed architectures, automated testing, observability, and performance optimization.

Education

University of Southern California, Los Angeles, CA **August 2024 – May 2026**
Master of Science in Computer Science *GPA: 3.75/4*
Coursework: Software Engineering, Analysis of Algorithms, Database Systems, Advanced NLP

Pandit Deendayal Energy University, Gandhinagar, India **November 2020 – May 2024**
Bachelor of Technology in Computer Science and Engineering *GPA: 3.88/4*
Coursework: Operating Systems, OOP, Software Engineering, Data Structures and Algorithms

Experience

USC Dornsife **Los Angeles, CA**
Software Engineer *September 2025 – Present*
- Architected a **distributed** Python platform with Azure SQL and Blob Storage for 500+ users, optimizing indexed queries and data access paths to reduce dashboard latency from 1.2 seconds to 350 milliseconds.
- Restructured fault-tolerant asynchronous workers processing 1,000+ jobs with explicit state transitions, error isolation, structured logging, and recovery paths, increasing concurrent throughput threefold while preventing cascading failures under production load.

Grad8 **Los Angeles, CA**
Software Engineer Intern *June 2025 – August 2025*
- Built **React** interfaces and Python Flask services handling 10,000+ daily requests, integrating secure file ingestion, REST APIs, and responsive workflows while maintaining stable performance during peak load.
- Developed a **multithreaded** batch-processing pipeline for 1,000+ files using ThreadPoolExecutor, validation checks, fault-tolerant execution, progress monitoring, repeatable error handling, and status visibility, reducing end-to-end processing turnaround by 60%.

USC AD Labs **Los Angeles, CA**
ML Research Assistant *January 2025 – May 2025*
- Built a Python **simulation pipeline** processing 100,000+ autonomous-driving trajectory logs, improving long-horizon planning by 18% through reproducible experiments, automated validation, performance tracing, and structured model configuration.
- Implemented spatial-temporal processing that combined HD-map geometry with vehicle trajectories through multi-head attention, improving closed-loop simulation stability 15% while supporting systematic debugging and comparative analysis across complex scenarios.

Aasma Technology Solutions **India**
Software Developer Engineer – SecureSync *August 2023 – July 2024*
- Engineered cross-platform software with automated executable builds and VM-based validation across Windows, Linux, and macOS, maintaining 20+ unit and integration test suites through Jenkins, PyTest, and Mockito.
- Optimized object-oriented integration modules, caching, pagination, and AWS data pipelines, improving response times 30%, increasing processing efficiency 68%, and reducing technical debt 20% through maintainable design patterns.

Aasma Technology Solutions **India**
Full Stack Developer Intern – VibrantChem *May 2023 – August 2023*
- Developed **React**, TypeScript, and Node.js capabilities across eight microservices, integrating WebSockets and REST APIs to synchronize more than two million records across real-time business workflows and modular service boundaries.
- Automated testing and containerized delivery with Jenkins, Docker, Jest, Mocha, Selenium, Sentry, and Azure monitoring, cutting deployment time 50% while improving release validation and production observability across environments.

Skills

Languages: Python, C++, Java, TypeScript, JavaScript, SQL, Bash, Shell, HTML, CSS
Frontend & Visualization: React.js, Next.js, Tailwind CSS, responsive UI, component architecture, state management, Gantt visualization
Backend & Systems: Flask, FastAPI, Node.js, Express.js, REST APIs, WebSockets, microservices, distributed systems, concurrency, multithreading, caching
Testing & Reliability: PyTest, Mockito, Jest, Mocha, Selenium, unit testing, integration testing, error handling, logging, monitoring, Sentry, performance optimization
Infrastructure & Practices: Git, Linux, Docker, Kubernetes, Jenkins, GitHub Actions, CI/CD, Azure, AWS, Agile, code reviews, technical documentation

Projects

CPU-Scheduling Simulator | Live | *C++, React, Operating Systems, EJ-2*
- Implemented six deterministic CPU-scheduling algorithms in **C++** with React controls and Gantt visualization, validating 1,000+ process combinations while reducing average turnaround time 25% and debugging time 20%.

Chat2SQL | Code | *Java, TypeScript, Docker, Maven, SQL*
- Built a cross-platform Java and TypeScript application supporting 10+ databases, schema-aware query generation, structured validation, automated testing, and Docker-based deployment across Windows, macOS, and Linux development environments.

FinNexus | Live | *TypeScript, WebSockets, Kubernetes, MongoDB*
- Built a real-time platform supporting 50,000+ users and one million transactions, reliably integrating WebSocket updates, failure monitoring, Kubernetes deployment, automated releases, and performance-tuned services under high-volume workloads.